Diamond Problem

```
class Animal {
    void makeSound() → Some Sound!
}

class Cat extends Animal {
    void makeSound() → "meow"
}

class Dog extends Animal {
    void makeSound() → "Woof"
}

class Cog extends Cat, Dog {
    void makeSound() → Ambiguity or Confusion
}
```

Not allowed in Java

Polymorphism :→

poly + morph
↓ ↓
many forms/shapes

Name	Place	Role
Sharan	Classroom	Student
	Restaurant	Customer
	Home	Son

only the role changes depending on the place. Person doesn't change!

The process by which the same entity/variable/method behaves differently under different scenarios, is called Polymorphism.

① (Compile-time) | Static | Overloading | Same Class

a) No. of parameters
b) Type of parameters

[It will not compile otherwise]

② Runtime | Dynamic | Overriding | Multiple Classes

C/C++/Java → decimal (double) (default)

Father — Building / Restaurant

Daughter — Building / Boutique @

Son — Building / Gym @

Overriding — RBI 6.3%

HDFC 7.2% | ICICI 8.1% | Canara 7.6%

(roi)

(Annotation)
@Override

* w3schools.com → C/C++/Java/Python/SQL
* Problems → Prepinsta Top 100 Codes (Logic)
* FreeCodeCamp → FreeCodeCamp.org
* Java → ① Kunal Kushwaha ② Telusko (Naveen Reddy)
  ③ Programming with Mosh (Mosh Hamedani)
  ④ Abdul Bari (Pseudo Code)
  ⑤ (Coders Arcade)